Lecture 01: AI/ML in Networking and Cybersecurity: The Landscape

NT120 – Introduction to AI for Computer Networks and Cybersecurity

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Course page: link.uit.edu.vn/NT120

Outline

1. Course Orientation
2. What Is AI, ML and Deep Learning
3. ML Systems Versus Traditional Software
4. Learning Paradigms
5. Applications in Networks and Cybersecurity
6. Why Security Is a Hard ML Domain
7. Checkpoint and Summary

Learning Outcomes

By the end of this lecture, students will be able to:

- Define artificial intelligence, machine learning and deep learning, and state how they relate
- Explain how an ML system differs from traditional rule-based software
- Distinguish supervised, unsupervised and reinforcement learning by the data each requires
- Recognise representative AI/ML applications in network management and cyber defence
- Explain why cybersecurity is a difficult application domain for machine learning

Course learning outcomes addressed: G1.1

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NT120 at a Glance

Structure

- 2 credits, 15 sessions of 90 minutes
- No separate laboratory credit: guided notebooks are done in class and at home
- Prior courses assumed: Introduction to Computer Networks, Introduction to Programming
- Minimum attendance: 70% of sessions

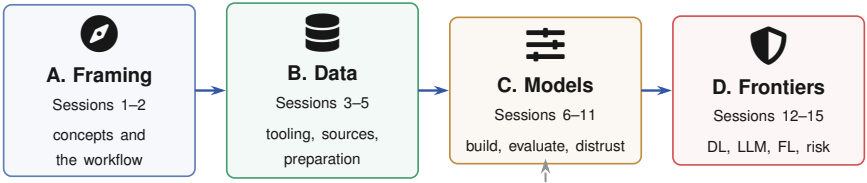
Assessment

- A1 continuous work 15%
- A2 midterm (Session 9) 35%
- A4 final examination 50%

Tools

- Python 3.10+, Google Colab
- Pandas, NumPy, Matplotlib, Scikit-learn
- Wireshark/tshark (demonstration only)

The Semester in Four Blocks



Every technique in Blocks B and C is anchored to a networking or security question. Block D is orientation: enough to read the literature and choose an advanced course, not enough to build the systems.

What This Course Is and Is Not

✔ It is

- Framing a network or security question as a learning task
- Handling flow records, logs and security events as data
- Training baseline models and reading their output honestly
- Knowing when the answer is “do not use ML here”

✘ It is not

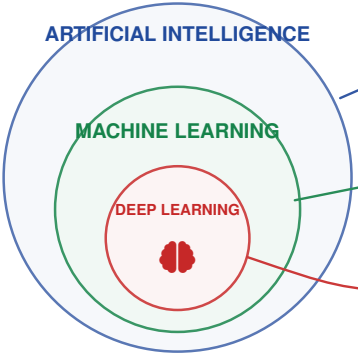
- A deep learning course: no architectures to implement
- An advanced network security course: no exploitation
- A mathematics course: intuition first, derivations later
- A tour of the newest models for their own sake

If you leave able to *frame* a problem and *distrust* a result, the course has done its job. Everything else is a technique you can look up.

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Three Nested Ideas



AI — any system performing tasks we associate with human intelligence, including hand-written rules and search

ML — the rule is *fitted from data* rather than written by a programmer; behaviour is induced, not specified

DL — ML with multi-layer neural networks that learn their own features directly from raw input

Saying It Precisely

- **Artificial intelligence** is a goal, not a technique. A firewall rule set written by an expert was called AI in 1985 and is not called AI today.
- **Machine learning** is a way of reaching that goal: give the program examples and an objective, and let an optimisation procedure produce the rule.
- **Deep learning** is machine learning where the model also learns the *representation*. You stop designing features and start designing architectures.

A useful test. Ask where the knowledge lives:

- In the source code, written by a person $\rightarrow$ rule-based system
- In parameters, fitted from data $\rightarrow$ machine learning
- In parameters that also encode the features $\rightarrow$ deep learning

"AI" in a product description tells you almost nothing. Ask which of the three the system actually is, and what data produced its parameters.

Five Eras of AI



Each era did not delete the previous one. Production security systems today still run large hand-written rule sets (Snort, YARA, SIEM correlation rules) alongside learned models. The interesting engineering question is where to put the boundary.

What Actually Changed

The algorithms behind today's systems are mostly old. Three other things moved.

Data

- Networks instrument themselves: flow exporters, logs, telemetry
- Labelled corpora became shareable
- Volume made statistical methods viable

Compute

- GPUs made large models trainable
- Cloud made the capacity rentable
- Edge accelerators moved inference to the device

Tooling

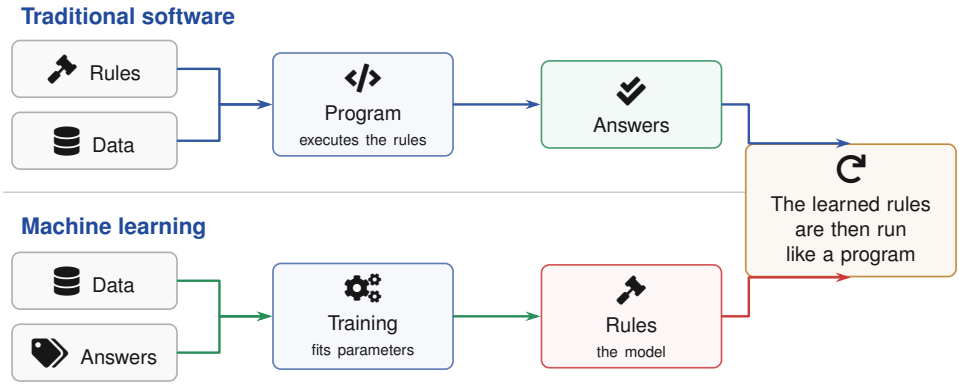
- Scikit-learn, PyTorch, TensorFlow
- Notebooks lowered the entry cost
- Pre-trained models became a starting point

This matters for you: the hard part of an AI project in a network operations team is almost never the algorithm. It is the data, the evaluation, and the integration.

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Two Ways to Build a Decision Rule



The arrows are reversed. In traditional software you supply the rules and get answers; in machine learning you supply answers and get the rules.

What Moves When Behaviour Is Induced

	Traditional software	**ML system**
Source of behaviour	Source code, written by a person	Parameters, fitted from data
Correctness	Absolute: the output is right or wrong	Statistical: right often enough
Testing	Assert expected outputs	Evaluate on held-out data
Debugging	Read and step through the code	Inspect the data, then the model
Change control	Code review, version control	Data, code *and* model versioning
Failure mode	Crashes, visible errors	Silent degradation as the world moves
Attack surface	Code and infrastructure	Code, infrastructure, *and the data*

The last two rows have no equivalent in ordinary software engineering. They are why Session 8 is about trusting results and Session 14 is about adversarial AI.

What This Costs You

- **Silent failure.** A rule that stops matching throws an error. A model that stops being right returns confident predictions that are wrong. Only monitoring reveals it.
- **Data-shaped bugs.** A mislabelled 2% of the training set produces a subtly wrong model that no code review will find.
- **Reproducibility.** The same code, a different random seed and a different data snapshot give a different system. Version all three.
- **Explaining a decision.** “Why was this host quarantined?” has an easy answer for a rule and a hard one for a model. In security, someone will ask.

⚠ Choose ML when the rule is genuinely hard to write and the data genuinely contains the pattern. Otherwise a rule is cheaper, faster, auditable, and does not drift.

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Three Paradigms, Three Data Requirements



SUPERVISED

You have: inputs *and* the correct label for each

You ask: what is the label of this new input?

Network example: is this flow benign or a DDoS attack?



UNSUPERVISED

You have: inputs only, no labels

You ask: what structure is here, and what does not fit?

Network example: which hosts behave unlike any other host?



REINFORCEMENT

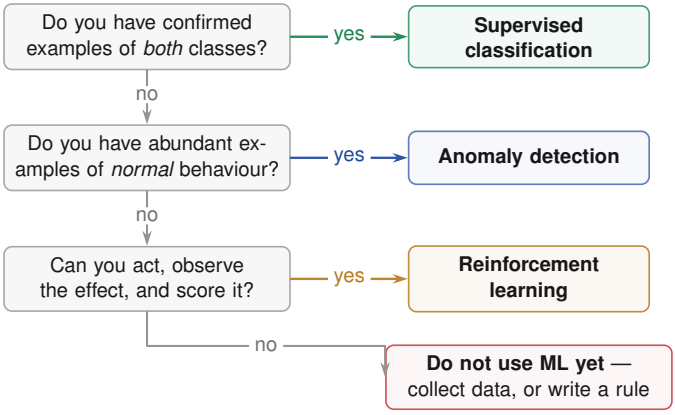
You have: an environment that returns a reward

You ask: which action should I take next?

Network example: how should routing adapt under load?

The paradigm is chosen by the data you can actually obtain, not by which method you find most interesting. In a security operations centre, labels are the scarce resource.

Choosing the Paradigm in Practice



Sessions 6–7 develop the first branch.

Session 10 develops the second.

Session 14 introduces the third.

Supervised and Unsupervised in Security, Side by Side

Supervised

- Needs a labelling process, which costs analyst time or a controlled capture
- Detects what it was trained on, and only that
- Gives a calibrated, comparable score per class
- Fails on genuinely new attack behaviour
- Examples: DDoS class in flow data, phishing e-mail, malware family

Unsupervised

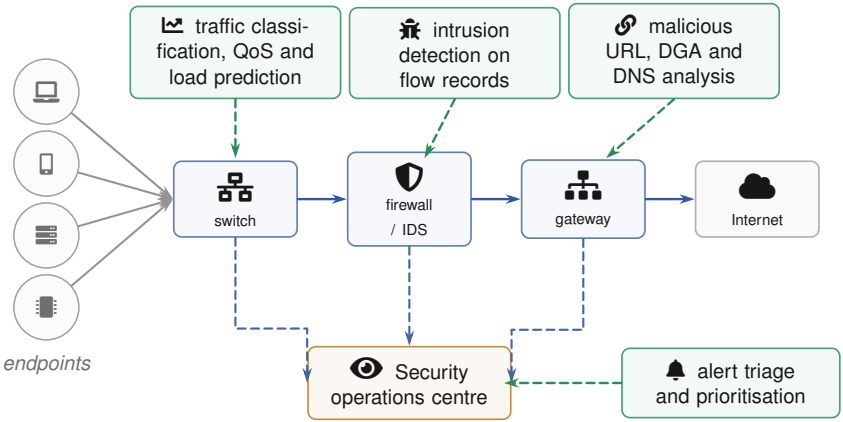
- Needs only normal traffic, which you already have
- Can surface behaviour nobody has named yet
- Gives a ranking, not a diagnosis: an analyst still has to look
- Generates false positives whenever “normal” shifts
- Examples: unusual host profile, rare beaconing pattern, insider activity

Most production security stacks use both: supervised models for the known families, an unsupervised layer for what the labels do not cover.

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Where ML Sits in a Network



Applications in Network Management

Traffic classification

- Identify the application behind an encrypted flow from timing and size statistics
- Drives QoS policy, capacity planning, shaping

Performance prediction

- Forecast congestion, latency, or link utilisation an hour ahead
- Turns a reactive operations team into a proactive one

Fault and failure prediction

- Predict device or link failure from telemetry and syslog patterns
- Schedules maintenance before the outage

Resource allocation

- Placement, scaling and routing decisions under changing demand
- The natural home of reinforcement learning (Session 14)

Notice that all four consume data the network already produces. No new instrumentation is needed to start.

Applications in Cyber Defence

Intrusion detection

- Classify flows or sessions as benign or attack
- Complements, and does not replace, signature rules

Malware and DGA detection

- Classify binaries, or domain names generated by an algorithm
- Catches variants that a signature misses

Phishing and spam

- Header, URL and content features; the oldest successful security ML application

Alert triage

- Rank and group the alert queue so the analyst sees the right ten first
- Often the highest-value application in a real SOC

In every one of these, the model is a *filter in front of a human*, not an autonomous decision maker. Keep that framing for the rest of the course.

Case Study 1 — Flow-Based Threat Detection in 5G

The problem. Modern mobile networks carry volumes that make per-packet deep inspection impossible: a deep packet inspection tool begins dropping packets in the low gigabits per second, and most traffic is encrypted anyway.

The approach (Gupta & Sheng, Chapter 1)

- Compute *payload-independent statistical features* from flows: durations, packet and byte counts per direction, inter-arrival statistics
- Train a deep model to classify flows, evaluated in a botnet detection setting
- Measure not only accuracy but *prediction runtime* across frameworks and traffic loads

Why it is a good teaching example

- The features survive encryption, which is the decisive practical constraint
- The paper treats latency as a first-class result, not an afterthought

Case Study 2 — DGA Domains and Malicious URLs

The problem. Blocklists and signature matching fail on a domain that has never been seen, which is exactly what a domain generation algorithm produces.

The approach (Gupta & Sheng, Chapter 6)

- Treat a domain name or URL as a *character sequence*, not a bag of engineered features
- Encode with character-level embedding, then let convolutional and recurrent layers extract the pattern
- A CNN–LSTM combination outperformed the classical classifiers in their comparison

Why it is a good teaching example

- It shows what deep learning buys you: the features are learned, not designed
- It uses DNS logs, a source every network already has, and we return to it in Session 12

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Four Structural Difficulties

The adversary adapts

In every other domain the data distribution is indifferent to your model. Here it actively moves away from it.

Extreme base rates

Attacks are a fraction of a percent of traffic. Accuracy becomes meaningless and precision collapses.

Asymmetric error costs

A missed intrusion and a false alarm are not comparable events, and the ratio between them differs per organisation.

Ground truth is expensive

Labels come from testbeds, blocklists or analysts. Each source is cheap, stale, or slow — pick one.

Sessions 7, 9 and 14 are, in effect, one long response to this slide.

The Base Rate Problem, Drawn



✗ Predictor A: "always benign"

Accuracy **99.6%** Attacks caught **0**

A model that learned nothing at all.

✓✓ Predictor B: a real detector

Accuracy **99.0%** Attacks caught **360**

Lower accuracy, and the one you want.

Predictor A beats Predictor B on accuracy by 0.6 points while detecting nothing. Any evaluation that reports a single accuracy figure would deploy the wrong system. Session 7 builds the metrics that separate them.

The Adversary Adapts

Ordinary machine learning assumes the data comes from a fixed, indifferent process. Security breaks that assumption in three ways.

- **Evasion.** The attacker modifies behaviour so it falls on the benign side of your boundary: padding, timing changes, tunnelling. Session 14.
- **Poisoning.** If your model retrains on data the attacker can influence, they can shape tomorrow's boundary today. Session 14.
- **Drift by competition.** Even without a targeted attack, tactics evolve. A detector trained on a 2017 corpus degrades on 2026 traffic for reasons no retraining schedule anticipated. Session 8.

⚠ Design question to carry through the course: *if an attacker read your feature list, what would they change first?*

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Checkpoint Questions

1. Give one networking task that is better solved by a hand-written rule than by a model, and say why.
2. Is DGA domain detection supervised or unsupervised? Describe what each version would require in terms of data.
3. An intrusion detection system reports 99.5% accuracy on a dataset in which 0.4% of flows are attacks. What is wrong with that report, and what would you ask for instead?

Discuss in pairs for three minutes, then five pairs report. Question 3 is a preview of the midterm.

Summary

- AI $\supset$ ML $\supset$ DL. Machine learning *induces* behaviour from data instead of specifying it, and deep learning also learns the representation.
- Reversing the arrows has consequences: statistical correctness, silent failure, data-shaped bugs, and the training data as part of the attack surface.
- The learning paradigm follows the data you can obtain. In security, labels are the scarce resource, which is why anomaly detection matters.
- AI is already routine in traffic analysis, intrusion detection, malware and DGA detection, phishing filtering and alert triage.
- Security is a hard ML domain for four structural reasons: adaptive adversaries, extreme base rates, asymmetric costs, and expensive ground truth.

Next session: turning a vague operational requirement such as “reduce phishing incidents” into a well-posed learning task, with a metric chosen before any model is trained.

Homework and Reading

Homework (assessed under A1)

- Complete the 10-question quiz on the course page before the next session
- Write five lines naming one AI application in networking or security that you have used or read about, and the data it must have consumed

Reading

- **[MLS]** Janapa Reddi, *Machine Learning Systems*, Ch. 1, §1.3–1.6 (pp. 6–21) and §1.9 (pp. 29–33); Ch. 2, §2.2–2.6 (pp. 56–76)
- **[GS]** Gupta & Sheng, *Machine Learning for Computer and Cyber Security*, Ch. 1 (introduction) and Ch. 6 (abstract and introduction)

Slides, notebooks, datasets and submission:
`link.uit.edu.vn/NT120`